

Threat Intelligence Types

1. Strategic Threat Intelligence

Strategic threat intelligence is high level information meant for senior management and executives rather than technical teams. It gives a broad overview of the current threat landscape, trends in cybercrime, and what types of attacks are targeting specific industries. For example a report saying that ransomware attacks against hospitals have increased by 40 percent in 2024 is strategic intelligence. It helps organizations make big decisions about security budgets and policies.

2. Tactical Threat Intelligence

Tactical threat intelligence focuses on the methods, techniques, and tools that attackers use to carry out their attacks. It is meant for security architects and IT teams who design defenses. For example information about how a specific hacking group uses phishing emails with fake PDF attachments to deliver malware is tactical intelligence. It helps security teams understand the attacker's behavior and build better defenses against those specific methods.

3. Operational Threat Intelligence

Operational threat intelligence provides real time details about active and ongoing attacks. It includes information about specific campaigns, who is behind them, what their motive is, and what targets they are going after. It is used by incident responders and SOC managers to prepare for and respond to active threats. For example knowing that a specific hacking group is currently running a campaign targeting banking institutions in Asia is operational intelligence.

4. Technical Threat Intelligence

Technical threat intelligence is the most specific and detailed type. It deals with actual technical evidence and indicators of compromise that can be directly used by security tools and SOC analysts during an investigation. This includes specific suspicious IP addresses, malicious domain names, file hashes of known malware, phishing email subject lines, and URLs used in attacks. For example knowing that the file hash d41d8cd98f00b204e9800998ecf8427e belongs to a known ransomware sample is technical intelligence. This type is directly fed into firewalls, SIEM systems, and antivirus tools to automatically detect and block threats in real time.

Investigating a Suspicious IP Address

When a suspicious IP address is found during a security investigation, the following steps should be taken:

Step 1: Check the IP on Threat Intelligence Platforms The first thing to do is look up the IP address on well known threat intelligence websites such as VirusTotal, AbuseIPDB, or Shodan. These platforms maintain databases of known malicious IP addresses reported by security

researchers and organizations worldwide. If the IP appears on these platforms it means it has a history of malicious activity.

Step 2: Perform a WHOIS Lookup A WHOIS lookup tells us who owns the IP address, which country it is registered in, and which internet service provider it belongs to. This helps identify whether the IP belongs to a legitimate company or a suspicious unknown source. Tools like whois.domaintools.com can be used for this.

Step 3: Check Geolocation Using a geolocation tool we can find out exactly which country and city the IP address is located in. If the IP is coming from a country that the organization has no business with it is a strong indicator of suspicious activity.

Step 4: Analyze the IP in SIEM Logs Search through the SIEM logs to find all activities associated with the suspicious IP address. Check what systems it tried to access, how many times it connected, what time the connections happened, and whether any login attempts were successful.

Step 5: Check Threat Feeds Many organizations subscribe to threat intelligence feeds that provide regularly updated lists of known malicious IP addresses. Checking the suspicious IP against these feeds can quickly confirm whether it is a known threat actor.

Step 6: Take Action Based on the findings block the IP address on the firewall if it is confirmed malicious, preserve all related logs as evidence, and document the findings in an incident report for future reference.

Real-life example: Investigating a Suspicious IP During a Data Breach Attempt

In 2023 a large e-commerce company's SOC team received an alert from their SIEM tool indicating that an unknown IP address 185.220.101.45 had been making repeated connection attempts to their database server outside of business hours at 2 AM.

The L1 analyst immediately escalated the alert to the L2 analyst who began a full IP investigation.

First the L2 analyst looked up the IP address on AbuseIPDB and discovered that this IP had been reported over 500 times by different organizations around the world for malicious activity including port scanning, brute force attacks, and data theft attempts.

Next they performed a WHOIS lookup and found that the IP was registered to an anonymous hosting provider in Eastern Europe commonly used by cybercriminals to hide their identity.

They then checked the geolocation and confirmed the IP was originating from a country that the company had absolutely no customers or business connections with making it even more suspicious.

After analyzing the SIEM logs they found that the IP had been scanning multiple ports on the database server and had attempted SQL injection attacks trying to extract customer payment data from the database.

The L2 analyst immediately blocked the IP address on the firewall, isolated the database server from the network temporarily, and escalated to the L3 expert. The L3 expert confirmed it was an attempted data breach, patched the SQL vulnerability the attacker was exploiting, added the IP range to the permanent blocklist, and filed a full incident report.

Thanks to proper threat intelligence tools and a structured investigation process the attack was stopped before any customer data was stolen.